PHILOSOPHY OF SCIENCE

Lecture 7: Laws of Nature

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OUTLINE

- Prehistory and concepts around laws of nature

- Contemporary accounts of laws of nature

PREHISTORY AND CONCEPTS AROUND LAWS OF NATURE



ORIGIN

- Where does the idea of laws of nature come from?

- Until the birth of modern science with Newton and Galileo, God was thought as directing the world

- This is prima facie incompatible with the existence of, e.g., equations of motion predicting the future on the basis of the present, deterministically

ORIGIN

- The response to this state of affairs was to say that equations of motion are laws of nature, and that laws of nature *govern* the evolution of the world

- God then is the entity fixing the laws of nature

- Problem solved: God is still in control even if we have, e.g., deterministic evolution

- Laws of nature provide the bridge!

ORIGIN

- The introduction of laws of nature however gives rise to a debate on their nature
- In particular, empiricists like David Hume are skeptical of the reality of laws: “No necessary connections between distinct entities”
- Opposed to Hume and Humeans, are those that defend the reality of laws
- Laws govern/encode necessary connections/have independent modal strength

PRELIMINARIES

- Before looking at laws of nature themselves, a clarification is in order
- The laws we will be interested in are laws of physics, and more specifically fundamental physics, so fundamental laws
- No special science laws for today
- This assumes that there is such a thing as fundamental laws, and that we can come to have sensible beliefs about them

PRELIMINARIES

- We currently have no candidate theory that could supply these laws (string theory?) and we don't even know if there is such a theory

- We will approach this task in an open-minded, suppositional way: we will ask ourselves, to the best of our understanding, what the fundamental laws might look like, given our understanding of current science and of the concept of laws of nature, while keeping in mind that this is ultimately subject to the whims of future science

- This is more problematic than it sounds for metaphysics, but for now we will leave this to the side

CONCEPTUAL CONNECTIONS

- Laws of nature are at the center of a broad nexus of philosophical concepts

- They are related to causation and counterfactuals, they are relevant to the interpretation and structure of physical theories, they are tightly connected to determinism and chance, and to notions of dynamics and evolution

- This web of concepts means that, by accepting an account of laws, we end up touching a variety of philosophical issues

CONCEPTUAL CONNECTIONS

- We tackle these one by one, starting from counterfactuals
- Recall what counterfactuals are: statements of the form "if A were to happen, then B would happen"
- These are, in philosophical jargon, modal statements: they concern the notions of possibility and necessity
- Equivalently, they are claims about possible worlds

CONCEPTUAL CONNECTIONS

- To see the connection with possible worlds, think of how you might establish the truth of a counterfactual

CONCEPTUAL CONNECTIONS

- To see the connection with possible worlds, think of how you might establish the truth of a counterfactual

- A first attempt is to claim that all worlds where A obtains also have B: too strong

- A better guess: the worlds where A and B obtain are more similar to ours than those where A obtains but B does not (Lewis, Stalnaker)

CONCEPTUAL CONNECTIONS

- What is the connection to laws? Laws support counterfactuals, since they determine a crucial respect of similarity between worlds

- In other words, “if it were A then B” is a true counterfactuals in case t is an instance of a law (if this match had been struck, it would light up)

- But it is not true in case it is not an instance of a law (if i had a coin in my pocket, it would be silver)

CONCEPTUAL CONNECTIONS

- Some, starting with Hume, and continuing with Lewis, have tried to exploit the connection between counterfactuals and laws to provide an account of causation

- In other words, two events are causally related if there is counterfactual dependence between them

- My striking of the match causes it to light because if I hadn't struck it, it wouldn't have light (counterfactual accounts of causation)

CONCEPTUAL CONNECTIONS

- The next point we are going to touch is the connection between laws and the structure and interpretation of scientific theories

- If we are realists, we can think of a scientific theory as specifying

  an ontology for the world, i.e. a collection of entities which correspond to the variables of the theory

  some dynamics for said variables, describing how they evolve: these are the laws

- A convenient way to represent especially the dynamics is via a state space, i.e. a space where every point is a state of the theory's entities, and a trajectory is the history of said entities

CONCEPTUAL CONNECTIONS

- Laws are then rules that select only some histories as allowed

- If you think of each history as a possible world, laws are a way to select which possible worlds are allowed: things that are true only in some of the law-allowed possible worlds are *nomologically possible*, while things true in all such worlds are *nomologically necessary*

- We call this nomological modality: the modality (recall, possibility and necessity, or possible worlds) associated with laws of nature

CONCEPTUAL CONNECTIONS

- The notion of nomological modality naturally leads to the notion of determinism
- What is determinism? It is, roughly speaking, the thesis that, if you know the state of the universe at one time and you know the laws of nature, you know its state at any other time
- In possible worlds/modal language, determinism means that given a segment of a nomologically possible world, there is only one full nomologically possible world with that segment as part
- Determinism looks highly tangled up with questions about laws of nature

CONCEPTUAL CONNECTIONS

- A classic example of a deterministic theory is Newtonian Mechanics
- General Relativity is a more interesting example, since not all models are deterministic
- Only *globally hyperbolic* models are deterministic, i.e. those models for which the state at one time plus the EFE determines the state at all times
- However, not all models of GR are globally hyperbolic: hence, whether GR is deterministic depends on whether we count non-globally hyperbolic models as physically significant or not

CONCEPTUAL CONNECTIONS

- Not all theories are deterministic however
- Quantum Mechanics, on some interpretations (collapse), is not deterministic
- If a theory fails to be deterministic, it involves chance in an ineliminable way
- In other words, we need probabilities to express the contents of the laws, and there is no way to get rid of them

CONCEPTUAL CONNECTIONS

- Another notion of determinism that Chen, motivated by Penrose, discusses is Strong Determinism, according to which there is only one world compatible with the laws

- Note that this is substantially stronger then the standard notion, since it implies that, according to theory *T*, there is only one world

- We do not have any real examples of this kind of determinism (string theory was hoped to have this property for a while, but it does not under any reasonable reading)

- Still, it can be conceptually interesting to draw this distinction, especially since we do not know ultimately what the fundamental laws will look like

CONCEPTUAL CONNECTIONS

- Another place where laws of nature play an important role is discussions over dynamics and the form of laws

- Some, like Tim Maudlin, have argued that laws of nature necessarily involve a notion of evolution in time, and associated notions of temporal asymmetry and dynamic evolution

- In particular, Maudlin thinks that laws produce later states of the world from earlier ones

- For Maudlin, laws of nature as *Fundamental Laws of Temporal Evolution*

CONCEPTUAL CONNECTIONS

- However, not everyone agrees with this characterization of laws

- Can you think of some examples/counterarguments?

CONCEPTUAL CONNECTIONS

- However, not everyone agrees with this characterization of laws
- Can you think of some examples/counterarguments?
- First of all, laws of physics for the most part are time symmetric, contra Maudlin, and temporal asymmetry is emergent, e.g., from thermodynamics
- Moreover, not all laws seem to involve dynamical evolution: some, like the EFE, are constraints

CONTEMPORARY ACCOUNTS OF LAWS OF NATURE



LAWS NOWADAYS

- Nowadays, debates around laws of nature mostly revolve around the Humean vs. non-Humean contrast

- Non-Humeans believe that laws govern the world, and that there is an irreducible form of nomological modality associated with the laws themselves

- Humeans on the other hand deny all of this: they do not think that law govern, and they think that nomological modal is reducible to non-modal notions

LAWS NOWADAYS

- A first question however might be: “why do we even need this?”
- For example, we might think that it’s enough to say that a law is a true universal generalization, i.e. a true statement of the form “all Fs are Gs”
- However, this doesn’t work, because there are accidental generalizations, i.e. true universal generalizations whose truth is completely accidental
- For example, the sentence “All gold spheres are less than one mile in diameter” happens to be a true universal generalization

LAWS NOWADAYS

- However, there is no particular reason, and in particular no law of nature, that forces this to be the case: it just happens to be the case

- Accounts of laws of nature try to capture the difference between genuine laws and accidental generalizations, by adding further conditions to a claim being a universally true generalization until we become convinced that only laws can satisfy those conditions

- In other words, accounts of laws of nature try to capture the sense in which laws force something to be true, rather than just happening to be true (as in the case of the gold spheres)

NON-HUMEAN VIEWS

- Non-Humean views, compared to Humean views, come in a larger variety of options

- There are disposinionalist views (what Chen calls Aristotelian reductionism), views based on universals (what Chen calls Platonist reductionism), or primitivist views (what Maudlin defends, and what Chen's minimal primitivism is a variant of)

- We will consider each in turn

MINIMAL PRIMITIVISM

- Chen's minimal primitivism is a primitivist view on laws of nature

- Primitive here means that we do not analyze a notion, in this case laws, in terms of some other notion, possibly more fundamental; the primitive notion is conceptually rock-bottom

- By saying that laws are primitive, we renounce to any possibility of analyzing them, in favor of avoiding the complicated metaphysical questions involved in any such analysis

- Whether this tradeoff is worth it is debatable

MINIMAL PRIMITIVISM

- Minimal primitivism, being a non-Humean view, is committed to laws governing the world

- The way this notion of governance is spelled out i the account is in terms of constraints: laws govern by constraining the space of possibilities admissible for the entities of the theory

- The analogy here is with things like mathematical axioms: axioms govern mathematics by fixing what is allowed and what is not, not by directing evolution in any reasonable way

MINIMAL PRIMITIVISM

- Insofar as this restriction is regarded as coming from laws and being irreducible, the view is genuinely non-Humean: laws do govern, and they govern by constraining

- Moreover, an advantage of this view over more heavy-weight primitivist views, like Maudlin's, is its minimalism

- We are committing ourselves to the minimal amount of irreducible nomological modality required to be non-Humean, and nothing else, since we only have the entities, and a constraint on how they are allowed to combine

- We do not place substantial restrictions on what laws are like Maudlin's requirement of FLOTEs

UNIVERSALS

- The second non-Humean account we will consider is the Dretske-Tooley-Armstrong account of laws

- This is an account according to which laws are relations of necessitation between universals

- "Necessarily, if F then G", where F and G are universals

- Once again, insofar as there is an irreducible modality coming from laws here, we are dealing with a non-Humean account

UNIVERSALS

- What is a universal? A universal is, roughly, the abstract entity that encodes a certain property in full generality

- For example, there is a universal for redness, such that every instance of the property red we see is just an instance that universal (so universals are, e.g., multilocated)

- The idea of the DTA account is that laws are necessary connections between such entities: for example, *F=ma* should be analyzed, for DTA, as "necessarily, if the universal F is such and such, then the universals m and a are such and such"

UNIVERSALS

- This account has the advantage of fitting rather naturally some simple examples of laws

- It is also a reductive account, in the sense that it analyzes the concept of law in terms of more fundamental concepts, i.e. universals; in this sense, it is more explanatory than the primitivist account, since it does explain what laws are

- However, it is unclear whether all laws fit into the DTA schema: for example, what about laws, like the EFE, that behave like constraints, or things like path integrals?

DISPOSITIONS

- The last non-Humean account we will look at is dispositionalism (what Chen calls aristotelian reductionism)

- This is an account where laws are analyzed in terms of causation, instead of universals

- In particular, in terms of a specific notion of causation based on something called a disposition

- A disposition is a property of the kind "glass breaks when it falls", i.e. a property that comes with a specific contextual clause that determines when it is realized

DISPOSITIONS

- One can analyze causal statements using dispositions, by saying that certain events have a disposition to happen when certain other events happen, where these latter events are their causes

- The stone caused the window to break because windows have a disposition to break when hit by stones

- We can then exploit the connection between causation and laws to recover laws, for example by saying that laws are the best possible summary of certain patterns of causal relations (Demarest)

DISPOSITIONS

- Is this account non-Humean? Yes, insofar as dispositions are understood as irreducible and modal

- Modality comes from the fact that dispositions encode a tendency, ther fact that under certain conditions something happens: in other words, that something is possible, and that under certain conditions it is actualized

- Since however the disposition is present also before actualization, it must be intrinsically modal

- Irreducibility is less obvious, but can be understood as the claim that dispositions should not be analyzed further: for example, in the glass case, one could think of explaining the disposition to break in terms of glass having the non-dispositional property of being fragile

- Friends of dispositions are committed to this kind of explanation not being allowed at the fundamental level

DISPOSITIONS

- Dispositional accounts however are reductionist when it comes to laws, since they analyze laws in terms of dispositions

- Hence, they have the same advantage as the DTA account over primitivism

- They also have similar issues: irreducible dispositions are mysterious, just like universals and primitive laws

- Moreover, it is not obvious that modern physical theories fit the dispositionalist mold

HUMEANISM

- It is now time to look at Humeanism, the main alternative to non-Humean accounts

- The modern version of Humeanism is chiefly due to David Lewis, and is based on the Best Systems Account of laws, also know as Mills-Ramsey-Lewis account

- According to the BSA, laws of nature are the axioms of a system that provides the best balance of strength and simplicity when accounting for the mosaic of spatiotemporally related local matters of fact of our world

- There is much to unpack here

HUMEANISM

- First of all, here we are reducing laws to something non-modal: spatiotemporally related local matters of fact

- Hence, laws are not anymore an irreducible feature of reality

- The spatiotemporally related local matters of fact are what is called the Humean mosaic: the bedrock ontological structure that determines the laws

- Laws are then distinguished from universal generalizations by virtue of encoding the best balance of strength and simplicity when it comes to encoding the content of the Humean mosaic

HUMEANISM

- Note that here we are reversing the relation between laws and world compared to non-Humean accounts: it is not the laws that govern/constrain/determine the world, but it is the world that determines the laws; this is the hallmark of the Humean account

- Two important advantages of the Humean account compared to non-Humean ones is that, first of all, we are not dealing with any mysterious entity or mysterious notion of governance

- Laws are effective summaries of the kinds of things that we are familiar with, i.e. local matters of fact, things like “a particle of mass m is here”

HUMEANISM

- Moreover, the Humean account has a natural connection to the laws the we discover within empirical science, since the balance procedure between competing summaries of the mosaic described by the BSA is an (idealized) version of what we do in actual science

- Non-Humean accounts, on the other hand, do not have a similar connection to empirical science: nothing stops the laws that govern from being different from the equations discovered by scientific theories

- However, Humeanism has problems of its own: most famously, not all physical theories seem to fit the template of laws being fixed by the mosaic (see quantum entanglement)

CONCLUSION

- We have seen a bird's eye view of laws of nature and their role in analyzing physical theories

- This is at its core a metaphysical debate, about the nature of laws itself; this is analyzed mostly in terms of a contrast between Humean and non-Humean approaches

- An important topic we have not touched is the status of laws in the special sciences